

# AKS Primality Test: Polynomial Congruence

**n = 7 (prime)**

$$\begin{aligned}(X + a)^7 \bmod 7 \\&= X^7 + a^7 \\&+ 7 \cdot (\dots) \equiv X^7 + a^7 \bmod 7\end{aligned}$$

□ **PASS**

Binomial coeffs:

$$\begin{aligned}C(7,1)=7, C(7,2)=21, \\C(7,3)=35, C(7,4)=35, \\C(7,5)=21, C(7,6)=7\end{aligned}$$

All divisible by 7!

**n = 9 (composite)**

$$\begin{aligned}(X + a)^9 \bmod 9 \\&= X^9 + a^9 \\&+ 9 \cdot (\dots) + C(9,3)X^6a^3 \\&+ C(9,2)X^7a^2\end{aligned}$$

□ **FAIL**

Binomial coeffs:

$$\begin{aligned}C(9,1)=9, C(9,2)=36, \\C(9,3)=84 \text{ NOT div by } 9, \\C(9,4)=126 \text{ NOT div by } 9, \\C(9,5)=126, C(9,6)=84, \\C(9,7)=36, C(9,8)=9\end{aligned}$$

## Key Insight

For prime  $p$ : all binomial coeffs  $C(p,k)$  are divisible by  $p$

For composite  $n$ : some  $C(n,k)$  are NOT divisible by  $n$

AKS: Check  $(X+a)^n \equiv X^n + a^n \bmod (X^r - 1, n)$  for small  $r$